

RTL Knowledge Graph Evaluation Report

1. KG Construction Criteria

The KG was built from the full merged ontology seed and merged label set, not from a 100-file sample.

Construction rules:

- Nodes: module, ip_block, label, port
- Edges: HAS_LABEL, HAS_PORT, INSTANTIATES
- Sources: out/merged_ontology_seed.jsonl and out/merged_labels.jsonl
- Projects: OpenTitan and Ibex
- Labels are inferred from module names, paths, ports, and instance clues

2. Full KG Summary

Modules: 1433

IP blocks: 38

Ports: 16766

Instance edges: 2103

Total nodes: 18251

Total edges: 22630

Projects: {'opentitan': 1142, 'ibex': 291}

Top labels: [['clocked', 1006], ['resettable', 966], ['hierarchical', 738], ['uart', 317], ['fifo', 220], ['spi', 154], ['crypto', 84], ['controller', 56], ['apb', 42], ['i2c', 30], ['ibex_core', 20], ['counter', 13], ['timer', 13], ['alu', 1]]

3. Baseline QA Benchmark

Question count: 150

Question types: {'exact_name': 50, 'label_plus_ports': 20, 'parent_child': 19, 'path_plus_port': 11, 'reverse_graph': 20, 'multi_hop': 15, 'semantic_bridge': 15}

Project mix: {'ibex': 50, 'opentitan': 100}

4. Baseline Retrieval Performance

baseline: hit@1=0.8067, hit@3=0.8467, mrr=0.8303, weighted_hit@1=0.7822, proxy_score=81.44

baseline by difficulty: {'easy': {'count': 50, 'hit_at_1': 1.0, 'hit_at_3': 1.0, 'mrr': 1.0}, 'hard': {'count': 50, 'hit_at_1': 0.78, 'hit_at_3': 0.9, 'mrr': 0.838}, 'medium': {'count': 50, 'hit_at_1': 0.64, 'hit_at_3': 0.64, 'mrr': 0.653}}

kg: hit@1=0.8467, hit@3=0.8667, mrr=0.8612, weighted_hit@1=0.8356, proxy_score=85.16

kg by difficulty: {'easy': {'count': 50, 'hit_at_1': 0.98, 'hit_at_3': 1.0, 'mrr': 0.99}, 'hard': {'count': 50, 'hit_at_1': 0.88, 'hit_at_3': 0.92, 'mrr': 0.9047}, 'medium': {'count': 50, 'hit_at_1': 0.68, 'hit_at_3': 0.68, 'mrr': 0.689}}

Retrieval metadata: {'modules_indexed': 1012, 'approved_labels': {'path': 'D:\\MyWork\\verilog\\out\\label_approval\\auto_approved_labels.jsonl', 'module_labels_added': 22, 'ip_context_labels_added': 115}, 'kg_version_tag': None}

5. Multi-Axis Question Set

Total questions: 175

Per level: {'L1': 35, 'L2': 35, 'L3': 35, 'L4': 35, 'L5': 35}

Per type: {'structure_understanding': 25, 'search_navigation': 25, 'comparison_similarity': 25, 'function_similarity': 25, 'generation_design': 25, 'code_explanation': 25, 'documentation_summary': 25}

Matrix counts:

- L1: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}
- L2: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}
- L3: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}
- L4: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}
- L5: {'structure_understanding': 5, 'search_navigation': 5, 'comparison_similarity': 5, 'function_similarity': 5, 'generation_design': 5, 'code_explanation': 5, 'documentation_summary': 5}

6. Multi-Axis Retrieval Performance

baseline: hit@1=0.8629, hit@3=0.8743, mrr=0.8824, weighted_hit@1=0.8601

By level:

L1: hit@1=0.8, hit@3=0.8286, mrr=0.8167, count=35

L2: hit@1=1.0, hit@3=1.0, mrr=1.0, count=35

L3: hit@1=0.8571, hit@3=0.8571, mrr=0.8929, count=35

L4: hit@1=0.7714, hit@3=0.7714, mrr=0.8071, count=35

L5: hit@1=0.8857, hit@3=0.9143, mrr=0.8952, count=35

By type:

code_explanation: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

comparison_similarity: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

documentation_summary: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

function_similarity: hit@1=0.52, hit@3=0.56, mrr=0.6433, count=25

generation_design: hit@1=0.96, hit@3=1.0, mrr=0.9733, count=25

search_navigation: hit@1=0.76, hit@3=0.76, mrr=0.76, count=25

structure_understanding: hit@1=0.8, hit@3=0.8, mrr=0.8, count=25

kg: hit@1=0.8686, hit@3=0.88, mrr=0.8733, weighted_hit@1=0.8683

By level:

L1: hit@1=0.8, hit@3=0.8571, mrr=0.8238, count=35

L2: hit@1=1.0, hit@3=1.0, mrr=1.0, count=35

L3: hit@1=0.8571, hit@3=0.8571, mrr=0.8571, count=35

L4: hit@1=0.7714, hit@3=0.7714, mrr=0.7714, count=35

L5: hit@1=0.9143, hit@3=0.9143, mrr=0.9143, count=35

By type:

code_explanation: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

comparison_similarity: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

documentation_summary: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

function_similarity: hit@1=0.56, hit@3=0.6, mrr=0.5733, count=25

generation_design: hit@1=1.0, hit@3=1.0, mrr=1.0, count=25

search_navigation: hit@1=0.72, hit@3=0.76, mrr=0.74, count=25

structure_understanding: hit@1=0.8, hit@3=0.8, mrr=0.8, count=25

Multi-axis metadata: {'modules_indexed': 1012, 'approved_labels': {'path':

'D:\\MyWork\\verilog\\out\\label_approval\\auto_approved_labels.jsonl', 'module_labels_added':

22, 'ip_context_labels_added': 115}, 'kg_version_tag': None}

7. Manticore Search Retrieval

This evaluation adds a Manticore Search-style BM25F full-text ranker over parser/LSP fields.

It also reports a hybrid variant that indexes KG labels, summaries, and reverse parent context.

manticore_parser_lsp: hit@1=0.8629, hit@3=0.8971, mrr=0.8935, weighted_hit@1=0.8423

manticore_hybrid: hit@1=0.8629, hit@3=0.8971, mrr=0.8781, weighted_hit@1=0.8423

Manticore metadata: {'modules_indexed': 1012, 'questions': 175, 'seed':

'D:\\MyWork\\verilog\\out\\merged_ontology_seed.jsonl', 'approved_labels': {'path':

'D:\\MyWork\\verilog\\out\\label_approval\\auto_approved_labels.jsonl', 'module_labels_added':

22, 'ip_context_labels_added': 115}, 'manticore_repo':

'D:\\MyWork\\verilog\\tools\\manticoresearch', 'manticore_source':

'https://github.com/manticoresoftware/manticoresearch', 'index_build_ms': 374.699,

'latency_ms': {'baseline': {'total_ms': 6908.985, 'avg_query_ms': 39.48}, 'kg': {'total_ms':

8685.073, 'avg_query_ms': 49.629}, 'manticore_parser_lsp': {'total_ms': 8190.132,

'avg_query_ms': 46.801}, 'manticore_hybrid': {'total_ms': 9410.487, 'avg_query_ms': 53.774}},

'note': 'Local BM25F model of Manticore Search over parser+LSP fields; schema/documents are emitted for real server loading.'}

8. VerilogEval Status

Official verilogeval runner/package was not available in this workspace.

All score_100 numbers are proxy scores, not official VerilogEval scores.